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COMMON QUARTERLY EXAMINATION - 2026

Standard XII

Reg.No. | | | | | | |

BIOLOGY

Part - I

Bio-Botany (35 Marks)

Time : 3.00 hrs *Marks : 70*

I. Choose the correct answer. 8 x 1 = 8

1. First cell of male gametophyte in Angiosperm is _____
   a) Microspore b) Megaspore c) Nucleus d) Primary Endosperm Nucleus
2. Phenotypic ration of dominant epistasis
   a) 9:3:4 b) 12:3:1 c) 9:7 d) 9:6:1
3. Changing the codon AGC and AGA represents
   a) Frameshift mutation b) Nonsense mutation
   c) Missense mutation d) Deletion mutation
4. In which technique is Ethidium Bromide used?
   a) Southern Blotting Technique b) Western Blotting Technique
   c) Polymerase Chain Reaction d) Agrose Gel Electroporosis
5. Virus free plants are developed from
   a) Meristem culture b) Organ culture c) Nucleus culture d) Cell culture
6. Trisomy was first reported in _____.
   a) Nicotiana b) Phyllium frondosum c) pisum d) Datura stramonium
7. Point mutation involves
   a) Deletion b) Insertion c) Change in a single base pair d) Duplication
8. For transformation, micro-particles coated with DNA to be bombarded with a gene gun are made up of _____.
   a) Silver (or) Platinum b) Platinum (or) Zinc
   c) Silicon (or) Platinum d) Gold (or) Tungsten

II. Answer any four questions. 4 x 2 = 8

9. What is Psychophily?
10. Define - Lethal allele.
11. What is the difference between missense and nonsense mutation?
12. Define multiple alleles. Give an example.
13. Write about Flavr-Savr Tomato.
14. What is Cybrid?

III. Answer any Three questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. List out the functions of Tapetum.
16. What are the reasons for Mendel's successes in his breeding experiment?
17. Give an account of Cryopreservation.
18. Write about 1. Totipotency 2. Differentiation 3. Redifferentiation.
19. Write about pBR 322 with neat diagram.

IV. Answer all the questions. 2 x 5 = 10

20. a) With a suitable diagram explain the structure of Megasporangium. *(OR)*
    b) Explain Dihybrid cross.
21. a) Compare the various types of Blotting Techniques. *(OR)*
    b) i) Write the advantages and disadvantages of Bt Cotton.
    ii) Define - Golden rice is the Biofortification.

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Part - II
Bio-Zoology (35 Marks)

I. Choose the correct answer. 8 x 1 = 8

1. The mode of reproduction in bacteria is by
   a) formation of gametes b) endospore formation
   c) conjugation d) zoospore formation
2. The androgen binding protein is produced by
   a) Laydig cells b) Hypothalamus c) Sertoli cells d) Pituitary gland
3. Select the incorrect action of hormonal contraceptive pills from the following.
   a) Inhibition of spermatogenesis
   b) Inhibition of ovulation
   c) Changes in cervical mucus impairing its ability to allow passage and transport of sperms
   d) Alteration in uterine endometrium to make it unsuitable for implantation
4. 'Universal Donor' and 'Universal Recipients' blood group are _____ and _____ respectively.
   a) AB, O b) O, AB c) A, B d) B, A
5. An operon is a
   a) Protein that suppresses gene expression
   b) Protein that accelerates gene expression
   c) Cluster of structural genes with related function
   d) Gene that switched other genes on or off
6. Which of the following rule is used to calculate the number of Barr bodies in a cell?
   a) N – 1 Rule b) Allen Rule c) N + 1 Rule d) Jordan Rule
7. Who proposed the Germplasm theory?
   a) Darwin b) August Weismann c) Lamarck d) Alfred Wallace
8. Spread of cancerous cells to distant sites is termed as
   a) metastasis b) Oncogenes c) Proto-oncogenes d) Malignant neoplasm

II. Answer any four questions. 4 x 2 = 8

9. How do you differentiate fission and fragmentation.
10. What do you mean by Ectopic pregnancy?
11. What is Mayer-Rokitansky syndrome?
12. What is Criss-cross inheritance?
13. What are Coacervates?
14. Write notes on Kala-azar.

III. Answer any Three questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. Draw the structure of human ovum and label the parts.
16. Differentiate Tubectomy and Vasectomy.
17. State any three goals of Human Genome Project.
18. Who disproved Lamark's theory of acquired characters? How?
19. Write the differences between cell mediated immunity and antibody mediated immunity.

IV. Answer all the questions. 2 x 5 = 10

20. a) Give a schematic representation of spermatogenesis and oogenesis in humans.

(OR)

b) Explain the genetic basis of ABO blood grouping in man.

21. a) Explain the structure of immunoglobulin with suitable diagram.

(OR)

b) What are the strategies to be implemented in India to attain total reproductive health.

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